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Department of Computer Science and Engineering
1st Year 2nd Semester B.Sc. (Engg.) Final Examination, Session: 2021-22
CSE 1204: Structured Programming Lab (1.5 Credits)

Time: 2 Hours

SET-A

Full Marks: 40

Question-1

Marks: 2

Write a program in C language that takes an integer n as input and prints "Leap Year" if n is a leap year otherwise prints "Not Leap Year".

Question-2

Marks: 3

Write a program in C language that takes a positive integer n as input and prints the sum of the digits of n . For example if $n = 3682$ then the output will be 19.

Question-3

Marks: 5 ✓

Write a program in C language that takes an integer n as input and prints a pattern of $2n-1$ number of lines as shown below.

$n = 4$	$n = 5$
<pre> ***** ***** *** * *** ***** ***** </pre>	<pre> ***** ***** ***** *** * *** ***** ***** ***** </pre>

Question-4

Marks: 5

Initialize an array of any number of integers *arr* in a program (C language). Now, assume another integer *x* and define a **recursive** function that returns the index of final occurrence of *x* in *arr* using **Binary Search**. If *x* is not found in *arr* then the function must return -1. For example if *arr* = {2, 3, 3, 3, 3, 3, 4, 4, 4, 4, 5, 5, 6} and *x* = 3 then the function will return 5.

Question-5

Marks: 5

Consider a matrix of integers of $m \times n$ dimension. Rows are indexed from 0 to $m-1$ and columns are indexed from 0 to $n-1$. An element of the matrix can have at most four neighbors (left, right, top, bottom). For example, the indices of neighbors of index-(2, 4) are:

- left = (2, 3)
- right = (2, 5)
- top = (1, 4)
- bottom = (3, 4)

	0	1	2	3	4	5	6	7
0								
1					top			
2				left		right		
3					bottom			

But, there may exist less than four neighbors for an element (border elements). For example: there exists three neighbors for index-(0, 3) and their indices are: left = (0, 2), right = (0, 4), bottom: (1, 3). Which means that the top neighbor doesn't exist because it goes out of bound. Similarly, there exists only two neighbors for index-(0, 0); none of the top and left neighbors exist.

Now, write a program in C language that takes two integers m and n in the first line denoting the dimension of a matrix. After that, the program accepts m lines of input containing n number of integers each representing the elements of the matrix. In the next line, the program takes two more integers r and c as input where r denotes the row-index and c denotes the column-index of an element in the matrix. The program should display the sum of the existing neighbors of the given element.

Sample Input Output

Input	Output	Clarification
4 5 2 8 1 3 0 9 7 5 2 1 1 8 3 5 2 7 5 9 9 2 2 3	16	Here the index of the given element is (2, 3). There exists four neighbors of the given element. They are: • left: 3 • right: 2 • top: 2 • bottom: 9 The sum of the neighbors is 16 which has been displayed as output.
3 4 1 4 2 9 3 3 3 3 7 6 4 8 2 2	17	Here the index of the given element is (2, 2). There exists three neighbors of the given element. They are: • left: 6 • right: 8 • top: 3 The sum of the neighbors is 17 which has been displayed as output. In this case, the bottom neighbor does not exist.

Question-6

Marks: 5

Write a program in C language that takes an integer n as input and prints all the prime numbers from 1 to n utilizing the method of **Sieve of Eratosthenes**.

Question-7

Marks: 6

Write a program in C language that takes two strings X and Y as input and determines whether

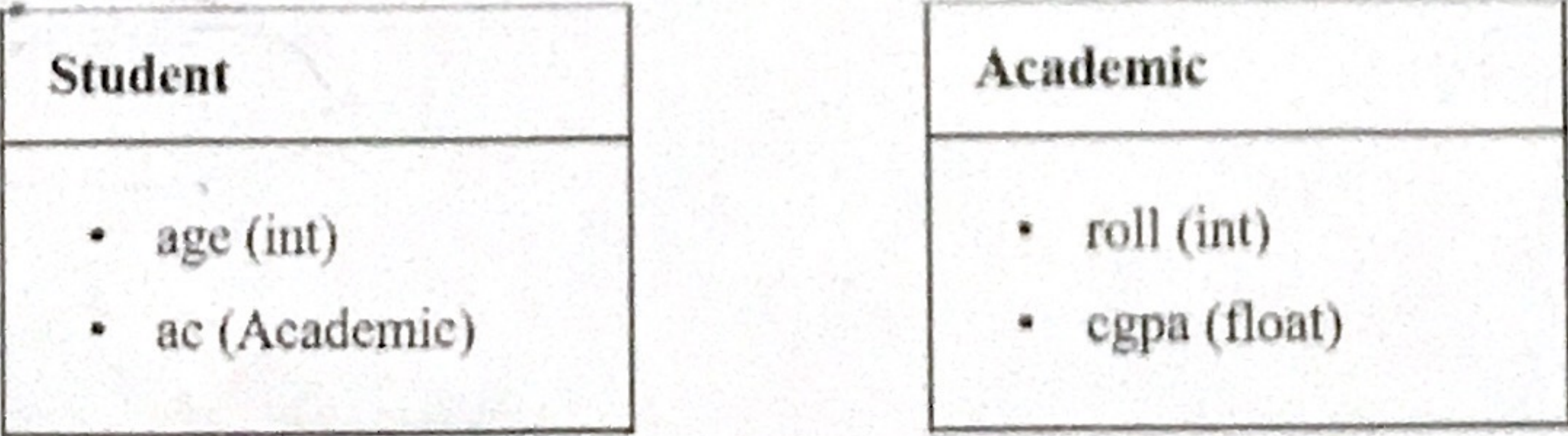
- Y is a substring of X or not
- Y is a subsequence of X or not
- Y is a prefix of X or not

For example, let $X = \text{"Bangladesh"}$ and $Y = \text{"Bangla"}$ then Y is the substring, subsequence and prefix of X . Again, if $X = \text{"Bangladesh"}$ and $Y = \text{"Bash"}$ then Y is not a substring or a prefix but a subsequence of X . If $X = \text{"Bangladesh"}$ $Y = \text{"Deb"}$ then Y is not a substring, subsequence, prefix of X .

Question-8

Marks: 9

Define two data types named "Student" and "Academic" as shown in the following diagram.



Now consider that, the information of some "Student" data types is written in a text file named "input.txt". Write a program in C language to sort the elements of "Student" data types based on the following criteria:

The array must be sorted by the descending order of the *cgpa* but if two *cgpa* are equal then the array must be sorted by the descending order of *age*. If there is a tie between two *age* then the array should be sorted by the ascending order of *roll*. It is ensured that the *roll* of each element in the array is unique.

You can consider that, the "input.txt" file is organized as follows:

- Each line in the file contains two integers and a float denoting the *roll*, *age* and *cgpa* of a Student respectively

The sorted information must be printed in the same file "input.txt", but the previous data must not disappear.

Additionally do not write the code for sorting in the main function. Write a separate function for sorting inside a header file named "sort.h". From your program only include that header file.

Sample Input Output

"input.txt"	"input.txt" after sorting	Remarks
202122104006 24 3.52 202122104005 27 3.50 202122104001 23 3.6 202122104003 24 3.52 202122104004 26 3.52 202122104008 22 3.6	202122104006 24 3.52 202122104005 27 3.50 202122104001 23 3.6 202122104003 24 3.52 202122104004 26 3.52 202122104008 22 3.6 After sorting: 202122104001 23 3.6 202122104008 22 3.6 202122104004 26 3.52 202122104003 24 3.52 202122104006 24 3.52 202122104005 28 3.50	Here the information after sorting has been written in the same file "input.txt" and previous data did not disappear.